Show all steps or explain how you found your answer.

1. Let $A = \{a, e, h, k\}$, $B = \{a, b, d, e, h, i, k, l\}$, and $C = \{a, c, e, m\}$. Find each of the following sets. No work necessary. (2 pts each)
   a. $A \cap B$
   $A \cap B = \{a, e, h, k\}$
   b. $A \cap B \cap C$
   $A \cap B \cap C = \{a, e\}$
   c. $A \cup B \cup C$
   $A \cup B \cup C = \{a, b, c, d, e, h, i, k, l, m\}$
   d. $A - B$
   $A - B = \emptyset$
   e. $A \cap (B - C)$
   $A \cap (B - C) = \{h, k\}$

2. Prove by showing that the left side is a subset of the right side and the right side is a subset of the left side. This is called a containment proof. (10 pts)

$$A - (B \cap C) = (A - B) \cup (A - C)$$

Let A, B, and C be sets. Recall that $x \in A - B$ iff $x \in A$ and $x \notin B$.

1: We will show $A - (B \cap C) \subseteq (A - B) \cup (A - C)$. Let $x \in A - (B \cap C)$, then $x \in A$ and $x \notin B \cap C$. So $x \notin B$ or $x \notin C$. If $x \notin B$, then $x \in A$ and $x \notin B$, hence $x \in A - B$. If $x \notin C$, then $x \in A$ and $x \notin C$, hence $x \in A - C$. In either case $x \in (A - B) \cup (A - C)$. Therefore, $A - (B \cap C) \subseteq (A - B) \cup (A - C)$.

2. We will show $(A - B) \cup (A - C) \subseteq A - (B \cap C)$. Let $x \in (A - B) \cup (A - C)$. Then $x \in A - B$ or $x \in A - C$. If $x \in A - B$, then $x \in A$ and $x \notin B$. Hence, $x \notin B \cap C$, so $x \in A - (B \cap C)$. If $x \in A - C$, then $x \in A$ and $x \notin C$. Hence, $x \notin B \cap C$, so $x \in A - (B \cap C)$. Thus, $x \in A - (B \cap C)$ in either case. Therefore, $(A - B) \cup (A - C) \subseteq A - (B \cap C)$.

$\therefore$ Since we have shown containment both ways, $A - (B \cap C) = (A - B) \cup (A - C)$. ■

3. Let $f(n) = 2\left\lfloor \frac{n}{2} \right\rfloor$. Tell whether f is a bijection from the set of integers to the set of integers. Check BOTH conditions and explain your reasoning for each. If the function is a bijection, find the inverse. *Hint - The function given contains the FLOOR function. (10 pts)

For any integer k, $f(2k) = 2\left\lfloor \frac{2k}{2} \right\rfloor = 2k$ and $f(2k+1) = 2\left\lfloor \frac{2k+1}{2} \right\rfloor = 2\left\lfloor k + \frac{1}{2} \right\rfloor = 2\lfloor k \rfloor = 2k$.
Thus, $f(2k) = f(2k+1)$ while $2k \neq 2k+1$. For example, $f(0) = 0 = f(1)$. Therefore, f is not injective.

For any $n \in \mathbb{Z}$, $f(n) = 2\left\lfloor \frac{n}{2} \right\rfloor$ is even, since it is 2 times an integer by definition of any even number. Therefore, there are no odd integers in the co-domain. Thus, f is not surjective from $\mathbb{Z} \to \mathbb{Z}$.

Since f is not injective or surjective, it is not a bijection from $\mathbb{Z} \to \mathbb{Z}$, so it has no inverse.

4. Find the closed-form expression for the recurrence relation $a_n = 2a_{n-1} - 3$ where $a_0 = -1$ using iteration. (10 pts)

$a_n = 2a_{n-1} - 3, \; a_0 = -1$

$a_1 = 2a_0 - 3 = 2(-1) - 3 = -5,$

$a_2 = 2a_1 - 3 = 2(a_0 - 3) - 3 = 4a_0 - 9,$ Pattern: $a_n = 2^n a_0 - 3(1 + 2 + 4 + \cdots + 2^{n-1})$.

$a_3 = 2a_2 - 3 = 2(4a_0 - 9) - 3 = 8a_0 - 21.$

Using the geometric sum $1 + 2 + \cdots + 2^{n-1} = 2^n - 1$:

$a_n = 2^n a_0 - 3(2^n - 1)$

$= 2^n(-1) - 3(2^n - 1)$

$= -2^n - 3(2^n) + 3$

$= 3 - 4(2^n)$

$= 3 - 2^{n+2}$.

$\boxed{\text{Closed form: } a_n = 3 - 2^{n+2}}$

$a_0 = -1$

$a_1 = 2(-1) - 3 = -5$

$a_2 = 2(-5) - 3 = -13$

$a_3 = 2(-13) - 3 = -29$

$a_0 = 3 - 2^{0+2} = 3 - 2^2 = -1$

$a_1 = 3 - 2^{1+2} = 3 - 2^3 = -5$

$a_2 = 3 - 2^{2+2} = 3 - 2^4 = -13$

5. Find the value of each sum using any SUMMATION FORMULA. You may, however, choose to use brute force on only ONE of the following. Be sure to show your work! (5 pts each)

a. $\sum_{i=1}^{8} 2i + 3 = 2\sum_{i=1}^{8} i + 3\sum_{i=1}^{8} 1$

$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \longrightarrow \sum_{i=1}^{8} i = \frac{8 \cdot 9}{2} = 36, \; \sum_{i=1}^{8} 1 = 8.$

Substitute and compute:

$2(36) + 3(8) = 72 + 24 = 96.$

b. $\sum_{i=4}^{10} 5i^2 = 5\sum_{i=1}^{10} i^2 = 5\left(\sum_{i=1}^{10} i^2 - \sum_{i=1}^{3} i^2\right)$

$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \longrightarrow \sum_{i=1}^{10} i^2 = \frac{10 \cdot 11 \cdot 21}{6} = 385, \; \sum_{i=1}^{3} i^2 = \frac{3 \cdot 4 \cdot 7}{6} = 14.$

Substitute and compute:

$5(385 - 14) = 1855.$

c. $\sum_{k=0}^{15} 4 \cdot 3^k$

$\sum_{k=0}^{15} 4 \cdot 3^k = 4\sum_{k=0}^{15} 3^k = 4 \cdot \frac{3^{16} - 1}{3 - 1} = 4 \cdot \frac{3^{16} - 1}{2} = 2(3^{16} - 1) = 2 \cdot 43{,}046{,}720 = 86{,}093{,}440.$

d. $\sum_{k=1}^{100} (-1)^k$

$\sum_{k=1}^{n} r^k = \frac{r(1 - r^n)}{1 - r}, \quad \sum_{k=1}^{100} (-1)^k = \frac{(-1)(1 - (-1)^{100})}{1 - (-1)} = \frac{(-1)(1 - 1)}{2} = 0.$

6. Let $A = \begin{bmatrix} 0 & 1 & 5 \\ 2 & 3 & -2 \end{bmatrix}$ *and* $B = \begin{bmatrix} 1 & -4 \\ 5 & 2 \\ 0 & -2 \end{bmatrix}$. Find each of the following, if possible. If not possible, explain why not. Show work! (5 pts each)

a. $AB = \begin{bmatrix} 0 & 1 & 5 \\ 2 & 3 & -2 \end{bmatrix} \begin{matrix} 0 & 2 \\ 1 & 3 \\ 5 & -2 \end{matrix} \begin{bmatrix} 1 & -4 \\ 5 & 2 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 0+5+0 & 0+2+-10 \\ 2+15+0 & -8+6+4 \end{bmatrix} = \begin{bmatrix} 5 & -8 \\ 17 & 2 \end{bmatrix}.$

b. $BA = \begin{bmatrix} 1 & -4 \\ 5 & 2 \\ 0 & -2 \end{bmatrix} \begin{matrix} 1 & 5 & 0 \\ -4 & 2 & -2 \end{matrix} \begin{bmatrix} 0 & 1 & 5 \\ 2 & 3 & -2 \end{bmatrix} = \begin{bmatrix} 0+-8 & 1+-12 & 5+8 \\ 0+4 & 5+6 & 25+-4 \\ 0+-4 & 0+-6 & 0+4 \end{bmatrix} = \begin{bmatrix} -8 & -11 & 13 \\ 4 & 11 & 21 \\ -4 & -6 & 4 \end{bmatrix}.$

c. $A + B$

Matrix addition requires the matrix be the same dimensions. Since A is 2×3 and B is 3×2, A+B is undefined.

7. Find the join, meet, and Boolean product of these two zero-one matrices. (5 pts each)

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

a. Join $A \vee B$ (no work necessary)

$A \vee B = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

b. Meet $A \wedge B$ (no work necessary)

$A \wedge B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

c. Boolean Product $A \odot B$ (show at least one intermediate step)

$A \odot B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{matrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{matrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} (1\wedge 0)\vee(0\wedge 1)\vee(0\wedge 0) & (1\wedge 1)\vee(0\wedge 1)\vee(0\wedge 0) & (1\wedge 0)\vee(0\wedge 1)\vee(0\wedge 1) \\ (0\wedge 0)\vee(1\wedge 1)\vee(1\wedge 0) & (0\wedge 1)\vee(1\wedge 1)\vee(1\wedge 0) & (0\wedge 0)\vee(1\wedge 1)\vee(1\wedge 1) \\ (1\wedge 0)\vee(1\wedge 1)\vee(1\wedge 0) & (1\wedge 1)\vee(1\wedge 1)\vee(1\wedge 0) & (1\wedge 0)\vee(1\wedge 1)\vee(1\wedge 1) \end{bmatrix}$

$= \begin{bmatrix} 0\vee 0\vee 0 & 1\vee 0\vee 0 & 0\vee 0\vee 0 \\ 0\vee 1\vee 0 & 0\vee 1\vee 0 & 0\vee 1\vee 1 \\ 0\vee 1\vee 0 & 1\vee 1\vee 0 & 0\vee 1\vee 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$

8. Describe an algorithm for finding the smallest integer in a finite sequence of integers. You may use pseudo code if you'd like, but it is not necessary. (10 pts)

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Procedure find smallest (a1, a2, ..., an: integers)
Min:= a1 ; i:=2
While i ≤ n
    if ai < min, then min:= ai
i:=i+1
return min {smallest value in the sequence}
```